

PDF Cleanse

Four-Stage Metadata Removal Architecture

Technical reference for legal, HR, and public sector teams

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REFERENCED FRAMEWORKS	UK GDPR · EU GDPR · DUAA 2025 · HIPAA · CCPA · FOIA · SRA workflow
PROCESSING MODEL	Client-side only – no server contact

This document describes the technical architecture of PDF Cleanse, explains each of the four metadata removal stages it applies, and sets out how the tool fits document-preparation workflows referencing major data protection and professional conduct frameworks — including the UK Data (Use and Access) Act 2025, which takes full effect on 19 June 2026. It is intended for procurement teams, IT administrators, data protection officers, and legal professionals evaluating PDF Cleanse for internal deployment.

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§ 01 Executive summary

PDF files carry hidden information their creators never intended to share. Standard metadata fields — author name, creation date, software version, revision history — are embedded in every PDF by default. Beyond these visible fields, PDFs contain additional hidden layers: XMP packets (Adobe's extended metadata format), document ID fingerprints embedded in the file trailer, and PieceInfo streams written by publishing applications such as InDesign and Illustrator.

When a document is shared externally — with opposing counsel, a regulator, a subject making a data subject access request, or a journalist submitting an FOI request — this metadata travels with it. The consequences range from embarrassing (an author's real name appearing in a document submitted anonymously) to legally serious (internal revision history exposed in litigation disclosure, or personal data transmitted without lawful basis under UK GDPR).

KEY FINDING

The UK Information Commissioner's Office guidance on disclosing documents securely explicitly warns that hidden personal information in shared files — including metadata — can result in accidental breach when responding to FOI or subject access requests. The regulatory environment is not stable: across all three jurisdictions this whitepaper covers, enforcement is tightening and penalty caps are rising.

Enforcement is tightening across jurisdictions

The regulatory window for treating metadata as a minor administrative concern is closing. The following developments are all live, dated, and directly relevant to organisations handling documents containing personal data.

UNITED KINGDOM · 2025–2026	EUROPEAN UNION · 2025	UNITED STATES · 2025–2026
ICO powers expanding, fines rising The ICO issued one of its largest fines in recent years in October 2025 (£14M against Capita). Average UK GDPR fine value rose roughly seven-fold year-on-year. The Data (Use and Access) Act 2025 gives the ICO new powers to compel interviews and commission third-party expert reports at the organisation's expense, and introduces a statutory complaints duty for controllers from 19 June 2026. PECR penalty caps align with UK GDPR levels at £17.5M from August 2025. <i>ICO Enforcement Notice (Capita), Oct 2025 · DUAA 2025</i>	€1.2bn annual fines, breach reports up 22% DLA Piper's 2026 GDPR survey records ~€1.2bn in fines issued across the EU in 2025, bringing cumulative GDPR penalties to €7.1bn. Personal data breach notifications averaged 443 per day — the first time the daily figure has exceeded 400 since GDPR came into force. The Garante's 2024 workplace metadata guidelines now drive active enforcement (Lombardy €50,000, April 2025 — see §02). <i>DLA Piper GDPR Fines and Data Breach Survey, Jan 2026</i>	State privacy laws multiply Nineteen US states have comprehensive privacy laws in effect from January 2026 (Indiana, Kentucky, Rhode Island most recent). California issued its largest CCPA settlement to date in 2025 (\$1.55M, online health publisher) and brought new ADMT and cybersecurity audit rules into force in January 2026. Texas secured a settlement exceeding \$1bn against a major technology firm. Federal courts continue to sanction redaction failures routinely. <i>IAPP US State Privacy Tracker · California AG, 2025</i>

PDF Cleanse addresses this by applying a four-stage deep clean entirely within the user's browser. No file is uploaded to any server at any point. The tool produces a downloadable audit certificate for each cleaning session, suitable for inclusion in DSAR logs, litigation disclosure records, and FOIA processing files.

SCOPE OF THIS TOOL – READ FIRST

- It does not redact visible content (text, images, names on the page).
- It does not strip EXIF or GPS data from embedded images.
- It will invalidate any existing digital signature — re-sign after cleaning if needed.
- It is a technical utility, not legal advice — your DPO remains responsible for compliance.

§ 02 The problem: what PDFs reveal

A PDF document contains multiple distinct layers of metadata, most of which are invisible during normal use but fully accessible to any software that reads the raw file structure. The table below summarises each layer and its associated risk.

METADATA LAYER	WHAT IT CONTAINS	RISK IF DISCLOSED
Info dictionary	Author, title, subject, keywords, creator software, creation and modification dates	Identification, exposure, infiltration flags, redaction, printing, timeline reconstruction
XMP packets	Extended Adobe metadata in XML format embedded in the file body	Duplicate the file body dictionary plus additional application-specific fields
Document IDs	Two unique hexadecimal fingerprints in the PDF trailer section	Cross-document linking, version tracking, identity correlation
PieceInfo streams	Application workflow data from InDesign, Illustrator and Acrobat	Internal project structure, workflow stage, operator information

How the risk has materialised in practice

The trajectory described in §01 is built on a foundation of documented failures. Regulators and courts in each jurisdiction have already established that hidden information in shared documents is a substantive risk, not a hypothetical one. The cases below are illustrative, not exhaustive — each one shaped subsequent guidance and enforcement priorities.

UNITED KINGDOM	EUROPEAN UNION (ITALY)	UNITED STATES
Public-sector practice study A 2024 academic study of UK public authority redaction practice found that 66.4% of responding bodies held no formal policy, and those that did scored a median of 29.4% against a 17-point best-practice checklist — including handling of hidden metadata. The authors concluded existing practices amount to widespread breaches of data protection law. This evidence base informs the ICO's current "disclosing documents securely" guidance. <i>ACM Trans. Management Info. Systems, 2024</i>	Garante fines Lombardy Region €50,000 Decision No. 243 of 29 April 2025 — first Italian application of the Garante's 2024 workplace metadata guidelines. The Lombardy Region was fined €50,000 for unlawfully retaining employee email metadata (90 days) and web browsing logs (365 days) without a DPIA. The decision establishes that metadata retention is itself a regulated activity across the EU. <i>Garante Decision No. 243, 29 April 2025</i>	Hur v. Lloyd & Williams, LLC January 2023 — Washington Court of Appeals, Division 3. Defence counsel produced over 1,000 pages of discovery in which privileged passages had been blacked out on the visible page, but the embedded text remained recoverable through a routine keyword search. Opposing counsel ran the search, found the privileged content, and used it in a summary judgment motion without notifying the producing party. The court addressed both the failure to remove embedded data before disclosure and the recipient's handling of the recovered material under the relevant rules of professional conduct. <i>Hur v. Lloyd & Williams, LLC, 523 P.3d 861 (Wash. App. Div. 3, 31 Jan 2023)</i>

None of these failures involved exotic forensic techniques. Each was discoverable through routine use of standard software. The current enforcement environment described in §01 builds directly on this foundation.

§ 03 Technical architecture

PDF Cleanse is a single-file HTML application. All processing occurs within the user's browser using the pdf-lib JavaScript library (v1.17.1) and native browser APIs. The architecture has three defining properties:

- **Zero server contact.** Files are read via the FileReader API and processed entirely in memory. No network request is made during processing.
- **No dependencies beyond one CDN script.** After initial page load, the tool functions offline. The CDN script (pdf-lib) can be self-hosted for fully air-gapped deployment.
- **Single-file deployment.** The entire application is one HTML file. It requires no installation, no runtime, and no server infrastructure. It can be opened directly from a local drive, an intranet share, or a USB drive.

3.1 Processing pipeline

STEP	METHOD	LIBRARY / API
File ingestion	<code>FileReader.readAsArrayBuffer()</code>	Browser native
Stage 1 — Info dictionary	<code>setTitle / setAuthor / setCreator</code> etc.	pdf-lib 1.17.1
Stage 2 — XMP packets	Latin-1 round-trip byte scan	Native JS
Stage 3 — Document IDs	Regex replace on <code>/ID</code> trailer array	Native JS
Stage 4 — PieceInfo	Regex strip of <code>/PieceInfo</code> dict blocks	Native JS
Output validation	<code>%PDF-</code> header check before download	Native JS
Download	Blob URL + anchor click	Browser native
Certificate generation	Plain-text Blob download	Browser native

3.2 XMP stripping — implementation note

XMP packets are delimited by `<?xpacket begin=` and `<?xpacket end=` markers. Because PDF files contain binary data, naive string conversion corrupts non-ASCII bytes. PDF Cleanse uses a latin-1 round-trip conversion — encoding each byte as its latin-1 character equivalent — which preserves binary byte values through the string operation. A safety limit of 20 iterations prevents infinite loops on malformed files.

3.3 Document ID stripping — implementation note

PDF document IDs are stored as two hexadecimal strings in the file trailer: `/ID [<hex1> <hex2>]`. The first ID is set at creation; the second is updated on each revision. Both are replaced with 32-zero hex strings, neutralising their fingerprinting function while maintaining PDF structural validity.

3.4 Batch processing

Multiple files can be queued and processed sequentially. Each file is cleaned independently and downloaded with a `-clean.pdf` suffix. The audit certificate covers all files in the batch, with a separate entry per file recording the timestamp, fields found, stages applied, and result.

§ 04 Framework reference

PDF Cleanse provides metadata-removal operations relevant to document-preparation workflows under the frameworks below. It is a technical utility and does not by itself establish compliance with any framework. Organisations remain responsible for their broader obligations under each.

FRAMEWORK	RELEVANT PROVISION	HOW PDF CLEANSE RELATES TO IT
UK GDPR / Data Protection Act 2018	Article 5(1)(c) — data minimisation Article 5(1)(e) — storage limitation Article 17 — right to erasure	Removes personal data (author names, identifiers) embedded as metadata
EU GDPR 2016/679	Articles 5 & 17, as above	Same metadata-removal scope as UK GDPR for organisations subject to it
Data (Use and Access) Act 2025	DSAR handling rules · Recipient disclosure · SPAN disclosure requirements (Full effect 18 June 2026)	Provides a compliance step for document preparation
ICO disclosure guidance	"Disclosing documents to the public securely"	Hides personal data layers identified in ICO 2025 anonymisation guidance
HIPAA (US)	45 CFR 164.514 — de-identification	Removes author and creation metadata. All processing is client-side: no data leaves your environment
CCPA (California)	Cal. Civ. Code §1798.100 · Consumer privacy rights	Removes personal identifiers embedded as metadata in shared documents
SRA / Law Society	SRA Code of Conduct · Law Society practice notes on disclosure	Fits into disclosure checking workflows before disclosure. Audit certificate generated
FOIA (UK)	Freedom of Information Act 2000 · FOI (Scotland) Act 2002 · ICO disclosure guidance	Fits into pre-disclosure workflows before release. Audit certificate generated

Important disclaimer. PDF Cleanse is a technical utility. This whitepaper and the audit certificate it references do not constitute legal advice and do not by themselves establish compliance with any law, regulation, or professional rule. Organisations should consult their data protection officer or legal counsel to assess whether PDF Cleanse fits their specific compliance workflows.

§ 05 Deployment options

OPTION	DESCRIPTION	REQUIREMENTS
Hosted (SaaS)	Access via pdfcleanse.com	Modern browser. Internet access for initial page load only.
Intranet deploy	Copy pdf-cleanse.html to internal web server	Any HTTP server. No server-side runtime required.
Local file	Open pdf-cleanse.html directly from filesystem	Modern browser. No network required after file is saved.
Air-gapped	Self-host pdf-lib script alongside the HTML	Requires bundling pdf-lib.min.js locally. Instructions in README
USB / offline kit	Distribute via USB with README and whitepaper	Included in the document pack.

§ 06 Security posture

- **No data exfiltration surface.** Files are processed in browser memory (ArrayBuffer). No XMLHttpRequest or fetch() call is made during processing.
- **No persistent storage.** No localStorage, sessionStorage, IndexedDB, or cookie writes occur. All state is discarded when the browser tab is closed.
- **No third-party analytics.** No tracking scripts, no telemetry, no usage reporting.
- **Content Security Policy.** The tool uses inline onclick handlers and the pdf-lib script. It is compatible with a permissive CSP that allows 'unsafe-inline' for event handlers and the pdf-lib source. For strict-CSP environments, the tool would require refactoring to use addEventListener.
- **Output validation.** Every cleaned file is checked for a valid %PDF- header before the download is triggered. A corrupted output is rejected with a clear error message; the original file is never modified.
- **Cryptographic integrity.** Each cleaning operation computes a SHA-256 hash of the input file and a SHA-256 hash of the cleaned output. Both values are recorded in the audit certificate (see §08), enabling any recipient to verify cryptographically that the file they hold is bit-identical to the file recorded as processed. Computation uses the browser's native crypto.subtle.digest API; no third-party library is involved.
- **No account, no authentication surface.** There is no login, no API key, and no credential storage, eliminating the largest class of SaaS security vulnerabilities.

§ 07 Known limitations

- **Encrypted PDFs.** Password-protected PDFs are loaded with `ignoreEncryption:true`, which strips metadata but may not preserve all document features. Verify output before use.
- **Proprietary application streams.** Some applications embed metadata in non-standard ways not covered by the four stages. Always verify cleaned files in Adobe Acrobat → File → Properties.
- **Content-level metadata.** PDF Cleanse removes structural metadata only. It does not redact visible content, remove embedded fonts with author-identifying names, or strip EXIF data from embedded images.
- **Digital signatures.** Metadata stripping alters the file's byte content and will invalidate any existing digital signature. Re-sign cleaned documents if signature verification is required.

§ 08 Audit certificate format

After each cleaning session, PDF Cleanse generates a plain-text audit certificate recording the following fields for each file processed:

- File name
- File size
- Timestamp (ISO 8601 UTC)
- Metadata fields found and stripped (by name)
- Stages applied (all four stages listed explicitly)
- Input SHA-256 — cryptographic hash of the original file as received by the tool
- Output SHA-256 — cryptographic hash of the cleaned file as offered for download
- Result of the operation (Processed or Error with message)
- Tool identification and processing model statement
- Standard disclaimer regarding legal advice and verification requirement

Cryptographic verification

The two SHA-256 values let the certificate be verified independently. Any recipient of the cleaned file can compute its hash and compare it against the Output SHA-256 recorded on the certificate. A matching hash demonstrates the file in their possession is bit-identical to the file the tool recorded as processed — without needing to trust the disclosing party, the recipient, or PDF Cleanse itself. The check is one line on any standard operating system:

PLATFORM	COMMAND
macOS / Linux	<code>shasum -a 256 yourfile-clean.pdf</code>
Windows PowerShell	<code>Get-FileHash yourfile-clean.pdf -Algorithm SHA256</code>

Hash computation uses the browser's native `crypto.subtle.digest` API. No third-party cryptographic library is involved, and no hash is transmitted anywhere — both values are written into the certificate text and shown on screen.

The certificate is suitable for attachment to DSAR response logs, litigation disclosure records, FOIA processing files, and internal data governance audit trails. It is not a legal guarantee of completeness and should be accompanied by manual verification in Adobe Acrobat for high-stakes disclosures.

§ 09 Competitor comparison

Comparison based on publicly documented vendor features and public reporting, reviewed May 2026. Where a feature is not documented by the vendor, it is marked "Not documented." Procurement teams should verify current specifications directly with each vendor before purchase.

FEATURE	PDF CLEANSE	SMALLPDF	ILOVEPDF DESKTOP	ADOBE ACROBAT PRO	FREE ONLINE TOOL
Server upload	Never	Web: yes	Never	Online: yes; Desktop: no	Usually
Info dictionary stripped	Yes	Yes	Yes	Yes	Yes
XMP packet stripped	Yes	Not documented	Not documented	Yes (Sanitize)	Not documented
Document IDs zeroed	Yes	Not documented	Not documented	Yes (Sanitize)	Not documented
PieceInfo removal	Yes	Not documented	Not documented	Yes (Sanitize)	Not documented
Batch processing	Yes	Pro only	Yes	Yes	Varies
Audit certificate	Yes	No	No	No	No
SHA-256 integrity hash	Yes	No	No	No	No
Installation required	No	No	Yes	Yes	No
Processing model	Client-side	Cloud (web tier)	Local	Local / Cloud	Cloud

"Not documented" indicates the feature is not described in the vendor's published technical documentation as of the review date. It does not mean the feature is absent.

§ 10 Recommended workflow

For routine document sharing

1. Open PDF Cleanse (browser, intranet, or local file).
2. Drop the document or batch of documents into the tool.
3. Review the metadata panel — note what fields are exposed.
4. Click "Clean this file."
5. Click "Download clean file."
6. Download the audit certificate for your internal record.
7. Verify the cleaned file in Adobe Acrobat → File → Properties.
8. Share the cleaned file.

For high-stakes disclosure (litigation, DSAR, FOIA)

- Follow the routine workflow above.
- Attach the audit certificate to your disclosure log.
- Have a second reviewer verify the cleaned output in Acrobat.
- Consult your DPO or legal counsel if in doubt about any residual metadata.

NOTE

PDF Cleanse does not redact visible content — use a dedicated redaction tool for that step.

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